INCH-POUND

MIL-C-70714 (AR)
AMENDMENT 7
24 March 1994
SUPERSEDING
AMENDMENT 6
26 December 1992

MILITARY SPECIFICATION

CARTRIDGE, 60MM, ILLUMINATING, M721
LOADING, ASSEMBLING AND PACKING

This amendment forms a part of MIL-C-70714 (AR), dated 28 August 1987, and is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Title: Add "M721" after "XM721"; and wherever XM721 appears in the specification.

PAGE 2

2.1.2, Government drawings: PRODUCT AND PACKAGING DRAWINGS: Change "XM776" to "M776"; and wherever XM776 appears in the specification.

Change title of 9354466 as follows:
"Container, Ammunition, Metal, PA124".

Change title of 9390711 as follows:

"Packing and Marking for Box, Wirebound, for Cartridge, 60MM, Illuminating, M721"

Inspection Equipment. . .

INSPECTION EQUIPMENT DRAWINGS: Delete "7297287 - Chamber Gage" and "7257965 - Limit Check Gage" and add "7256348 - Alignment Gage."

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PAGE 3

3.2.1, Add new paragraph as follows:

"3.2.1 Tail cone. The inside surface of the tail cone must be free of dimples after drilling holes and staking of the shear pin. A dimple is considered to be a raised surface exceeding 0.010 inches high."

PAGE 4

3.5.6, Fuze times: Change "19.0" to "21.0" (2 places) and "26.0" to "27.0" (2 places).

PAGE 6

Table II, Examination or test: Change "Fast" to "First" and change under cartridge examination, the sample size (2 places) from "50" to "176."

PAGE 8

4.4.2.1, Major 107: Delete in its entirety.

Add new Major 108 as follows:

"108 Compression pad(s) missing or improperly placed ...0.40% ...3.2 ...Visual."

PAGE 9

4.4.2.2, Major 103: Delete in its entirety.

PAGE 10

4.4.2.3, Major 106: Change "Visual" to “Gage”.

Delete Major 101 and 102 in their entirety.

Add new Majors 108 and log as follows:

"108 Tail cone improperly assembled to the body tube assembly ... (a) ...3.2.1 ...Visual/Gage.

109 Presence of dimple on inside surface of tail cone ...(a) ...3.2.1 . . .Visual/Gage."

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PAGE 10 (Cont'd)

Change Note (a) to read as follows:

"(a) Defect callout and alternate sampling plan are depicted in 4.4.3.4."

PAGE 12

4.4.2.5, Major 101: Delete inspection method "Gage" and substitute "7256348."

Major 108: Delete inspection method "Visual" and substitute "Gage."

Major 109 and 110 and Note 1: Delete in their entirety.

PAGE 13

* 4.4.2.5, Under PROVING GROUND TEST, change sample units from "50" to "96/50."

PAGE 14

4.4.2.6, Add Major 103 as follows:

"103 Inner tube missing . ..0.40% . ..5.1 ...Visual."

PAGE 15

4.4.2.7, Under Minor 201, change "5.1" to "5.3."

PAGE 16

4.4.2.8, Title: Change "Contained" to "Container."

Delete Major 104 in its entirety.

Add new Minor as follows:

"201 Graphical firing scale missing . . .0.65% ...5.2 . . .Visual."

PAGE 17

4.4.2.9, Title: Change "Contained" to "Container."

Under Major 101, change "5.2" to "5.3."

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4.4.3.2.1.1, Change "85" to "200."

4.4.3.3, Last sentence: Delete end of sentence as follows:

"made on the applicable ... the rejected production."

4.4.3.4, Delete in its entirety and substitute the following:

"4.4.3.4 Alternate sampling plan for examination of depth, diameter of shear pin holes and presence of dimples. The depth and diameter of the shear pin holes and presence of dimples shall be inspected in the following manner using approved gages. Beginning with the first production shift, ten (10) consecutive drilled inert units shall be inspected and found defect free prior to rate production. Then inspect five (5) units every two (2) hours until the end of the shift. Five (5) units shall be inspected at the end of the shift. If a defect is found in any of the five (5) units, the machine set up shall be adjusted until ten (10) consecutive units are defect free.

4.4.3.4.1 Tail cone and body tube inspection. The projectile loading assemblies shall be subjected to a load (pull) to shear off the pins; slowly remove the body tube assembly from the tail cone and once the canister is visible, hold the canister down to prevent movement from the tail cone while removing the body tube. Inspect the cable by lifting candle slightly, if one coil of the cable is above the cable protectors, the sample shall be classed defective. Release the parachute expulsion spring; if the parachute folding and packing assembly does not pop out freely, the sample shall be classed as defective. If the drogue chute is tangled or improperly packed, the sample shall be classed as defective. If the cable is frayed, kinked or deformed, the sample shall be classed as defective. Inspect the inside surface of the tail cone where the shear pin holes were drilled for "dimpling." If the presence of two dimples 180 degrees apart are shown, the sample shall be classed as defective.

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4.4.3.4.2 Product shift examination. Beginning with the first production shift the (10) consecutive projectile assemblies shall be inspected and found defect free prior to rate production. Then inspect five (5) units randomly selected for every two (2) hours until the end of the shift. If a defective unit is found in any group of five (5), the production represented by the units shall be rejected and the machine set up shall be adjusted until ten (10) consecutive units are defect free.

4.4.3.4.3 Alternate sampling plan procedure. The alternate sampling plan procedure depicted in 4.4.3.4.2 shall be applicable to each drilling machine/equipment setup.

4.4.3.4.4 Tooling change. A change of tooling will require a first piece inspection prior to continuing production. The sampling plan will be the same as that prior to the tooling change.n

* 4.4.3.5, Delete first sentence in its entirety and substitute the following:

"a. First Article test. One hundred and seventy-six (176) sample cartridges shall be selected for first article testing. The cartridges shall be separated into three groups: one group (48 samples) conditioned for 24 hours at plus 145 ± 2.5°F, one group (80 samples) conditioned for 24 hours at plus 70 ± 2.5°F, and one group (48 samples) conditioned for 24 hours at minus 50 ± 2.5°F. Each group shall be separated into equal subgroups; one subgroup tested at Charge 1 and one subgroup tested at Charge 4.

b. Initial production test. Beginning with the first lot produced and continuing until three consecutive lots have successfully passed the acceptance criteria specified herein, ninety-six (96) sample cartridges shall be selected for this test. The cartridges shall be separated into three groups: one group (34 samples) conditioned for 24 hours at plus 145 ± 2.5°F, one group (28 samples) conditioned for 24 hours at plus 70 ± 2.5°F, and one group (34 samples) conditioned for 24 hours at minus 50 ± 2.5°F. Each group shall be separated into equal subgroups; one subgroup tested at Charge 1 and one subgroup tested at Charge 4.

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c. Production test. After successfully completing the initial production test, the next ten consecutive production lots (50 samples) shall be conditioned for 24 hours at plus 70 ± 2.5°F. The next production lot (eleventh lot) shall be tested and evaluated in accordance with the initial production test method of paragraph 4.4.3.5.b (96 samples). Upon successful completion of the eleventh lot, the next production lot shall be counted as the beginning of another ten consecutive lots (50 samples). Again, the eleventh production lot shall be tested and evaluated in accordance with the initial production test method (96 samples) and so on. Upon unsuccessful completion of any eleventh lot, the ballistic test shall revert to the requirements of paragraph 4.4.3.5.b."

* 4.4.3.5, Table, Categories, change from "Major: Illuminant fails to ignite or provide effective illumination for 32 seconds or less (See Note 2)" to "Major; Illuminant fails to ignite or provides effective illumination for less than 32 seconds in the air. (See Note 2 & Note 3)."

Note 2, Delete in its entirety and substitute the following:

"Note 2: If the fuze performance of any round in the sample meets the fuze functioning time requirements within plus or minus one second of the fuze settings specified, the mortar elevation (QE) is proper and the "effective illumination" time is less than thirty-two seconds, the performance of the cartridge will be considered valid for evaluating the acceptability of the lot for illumination time defects. The following are considered some causes of this defect:

a. Streaming chute/ripped chute.
b. partial or fully tangled chute.
c. Chute fails to deploy.
d. Delayed deployment of the parachute.
e. Illuminant detached from canister (illuminant dropout).
f. Parachute detached from canister.

Note 3, Fuze functioning time is defined as the elapsed time from round initiation in the mortar tube to first flash as fuze ejects the payload."

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4.4.3.5.1, Delete in its entirety and substitute the following:

"4.4.3.5.1 Overall acceptance criteria. In addition to the individual accept/reject criteria of 4.4.3.5, the production lot sample shall be rejected if a total of three or more major defects occur. The initial production and any eleventh lot (96 samples) shall be rejected if a total of four or more major defects occur. In the event that neither a misfire nor a fuze non-function/non-separation defect occurs during testing, the initial production and any eleventh lot allows a maximum of three (3) major defects for illumination less than 32 seconds in the air. The first article sample shall be rejected if a total of six or more major defects occur. In the event that neither a misfire nor a fuze non-function/non-separation defect occurs during testing, the first article allows a maximum of five major defects for illumination less than 32 seconds in the air. However, if either one misfire or one non-function/non-separation defect occurs, the first article allows a maximum of four major defects for illumination less than 32 seconds in the air. If one misfire and one non-function/non-separation defect occur, the first article allows a maximum of three (3) major defects for illumination less than 32 seconds. If either two or more misfires or non-function/non-separation defects occur during testing, the first article shall be rejected."

4.4.3.6, Delete in its entirety.

PAGE 24

4.4.5.7, Line 19: Delete sentence starting with "the cartridges shall be conditioned. .."

Test settings: Change "19.0" to "21.0" and "26.0" to "27.0."

Add to paragraph as follows:

"Meteorological data should be checked prior to and during testing. If the wind conditions exceed the TECOM firing table TOP thresholds or following criteria, then the test should be halted until the conditions are favorable.

a. When firing to maximum ordinates below 1,500 meters:

1. Surface winds exceed 10 knots.
2. Gust winds exceed 5 knots.

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b. When firing to maximum ordinates above 1,500 meters:

1. Surface winds exceed 20 knots.
2. If the wind speed changes by more than 15 knots or the wind direction changes by 800 mils when compared to two successive rawinsondes at a constant height. If the directional change exceeds 800 mils but the wind speed is less than 5 knots at the comparison height, firing may continue."

PAGE 25

4.4.5.8, Delete in its entirety.

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